



The Impact of Poverty and Food Access on Adolescent Obesity

Youth-Led Public Health Innovation

By Hitha Ganganala



Our Mission

To reduce adolescent obesity by addressing poverty-related barriers to nutritious food access through data-driven research, education, and policy-focused solutions that promote health equity.

Statement

Member



**Hitha
Ganganala**

Hitha Ganganala is a high school student at Wichita Collegiate School with a strong interest in public health, youth advocacy, and civic leadership. She leads student-driven public health research focused on adolescent health issues, including obesity and food access, using surveys, local data, and state and national datasets to identify disparities and inform solutions. She is active in HOSA and DECA, a varsity and USTA tennis player, and a committed volunteer. Hitha is focused on turning youth research into data-informed civic impact.



Progress



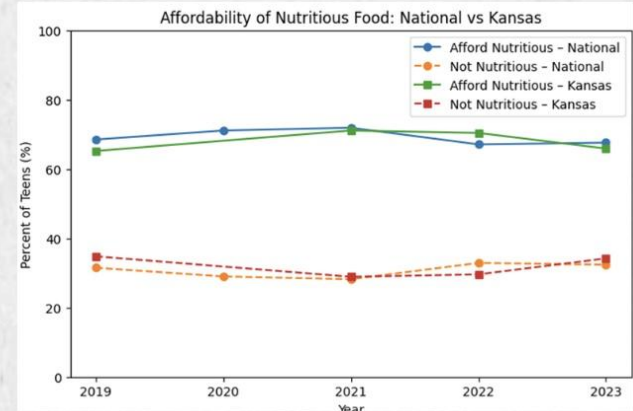
- Led a youth-driven public health research project on how poverty and food access impact adolescent obesity.
- Surveyed 135 Wichita-area high school students and analyzed Kansas and national data to identify income-based health disparities.
- Identified key barriers for low-income youth and developed evidence-based policy and community recommendations, presented through data visualizations and a formal research poster.



Issue/Problems

- Adolescent obesity is a growing public health crisis
- Disproportionately affects low-income youth
- Driven not just by behavior, but by access, affordability, and environment
- Disproportionately affects low-income youth

**Affordability of Nutritious Food:
National vs Kansas (2019–2023)**



Trends in adolescents' ability to afford nutritious food compared with reliance on non-nutritious food, nationally and in Kansas (2019–2023). Kansas data for 2020 were unavailable due to COVID-19 survey disruptions.

Why this is a civic problem?

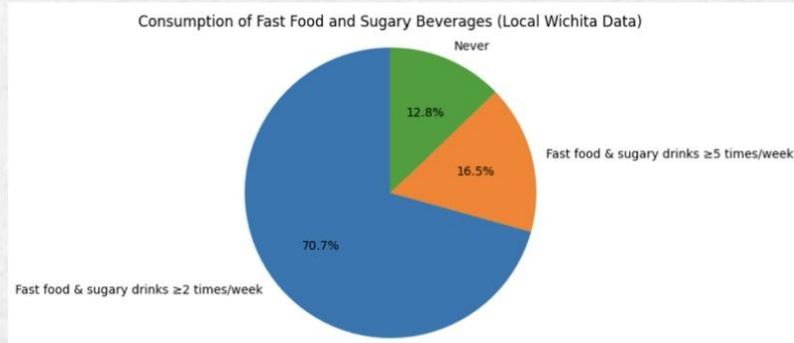
- Health outcomes are shaped by systems, not just choices
- Food access depends on:
 - Income
 - Neighborhood resources
 - School meal programs
- This creates inequity across communities
- Public policy and community action can change outcomes





Findings

Consumption of Fast Food and Sugary Beverages Among Wichita Adolescents



The majority of surveyed Wichita adolescents report frequent consumption of fast food and sugary beverages, highlighting a significant dietary risk factor associated with obesity.

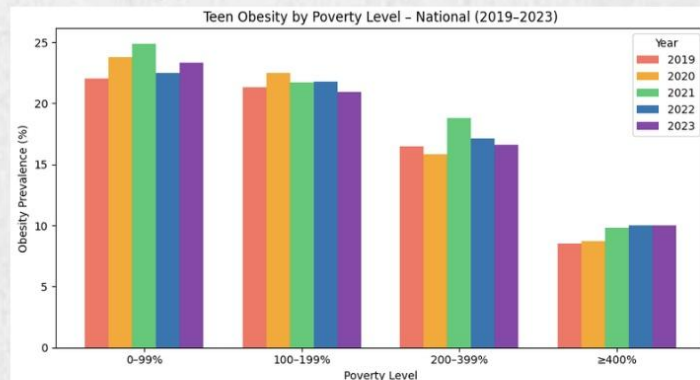
- Obesity rates higher among low-income adolescents
- Many students report:
 - Reliance on fast food and sugary beverages
 - Limited access to affordable nutritious meals
- Time and cost are major barriers to healthy eating



Relevance

- Low-income adolescents face consistently higher obesity rates, both nationally and in Kansas, with disparities persisting from 2019–2023.
- Food affordability is declining, increasing reliance on fast food and sugary beverages among teens.
- Physical activity levels remain below recommended guidelines, compounding diet-related risks.
- Adolescent obesity leads to long-term health consequences, making early intervention critical.

National Adolescent Obesity Trends by Poverty Level (2019–2023)



Across 2019–2023, obesity is highest in the 0–99% poverty group and decreases stepwise as income rises, with the ≥400% group consistently lowest. Despite year-to-year fluctuations, the income gap remains stable, highlighting a persistent disparity.



Limitations

- Awareness is high, but access is limited—students often know what's healthy but can't afford or access it.
- Existing programs are underutilized or inconsistently supported, especially in schools.
- Cost and time remain major barriers to healthy eating for low-income families.
- Youth voices are rarely included in designing prevention strategies, reducing relevance and impact.



Our Solution



A youth-led, data-driven public health model that addresses the root causes of adolescent obesity by improving access, affordability, and engagement.

- Improve nutrition access in schools by supporting affordable, healthy meal options for low-income students
- Targeted nutrition education focused on practical, low-cost food choices for families
- Increase opportunities for physical activity through school and community partnerships
- Youth-centered design, using student data and feedback to guide programs and ensure relevance

Focus: prevention, equity, and scalable community impact



Funding Allocation

Funding will allow us to move from research to real-world implementation.

- Pilot program development
 - Design and launch a small-scale school or community pilot
- Partnership building
 - Work with schools, nonprofits, and food access organizations
- Program materials & outreach
 - Educational resources, toolkits, and community engagement
- Data collection & evaluation
 - Measure participation, access, and health-related outcomes
- Youth leadership support
 - Training, mentorship, and coordination to sustain the project

Goal: build a tested, scalable model that can be expanded to additional schools and communities.





Impact

Near-Term Impact

- Increased access to affordable, nutritious food for low-income students through school and community partnerships
- Improved student awareness of realistic, low-cost healthy choices, not just ideal recommendations
- Greater student engagement in health programs due to youth-led design
- Stronger school community collaboration around nutrition and physical activity

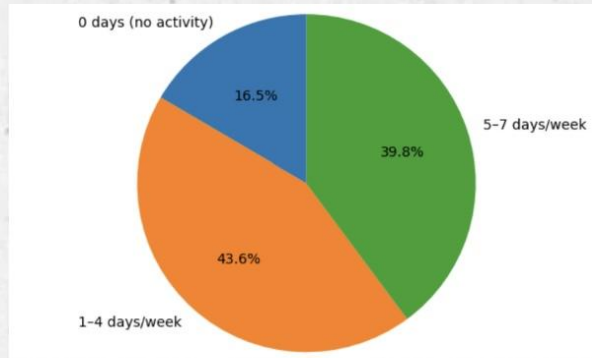
Long-Term Impact

- Reduction in adolescent obesity risk, particularly among low-income populations
- Narrowing of health disparities driven by poverty and food access
- Improved lifelong health outcomes, including lower risk of diabetes, heart disease, and mental health challenges
- A scalable, evidence-based prevention model that can be adopted by other schools and communities
- Sustained youth leadership in public health, empowering teens to shape policies and programs that affect them



Our Evidence

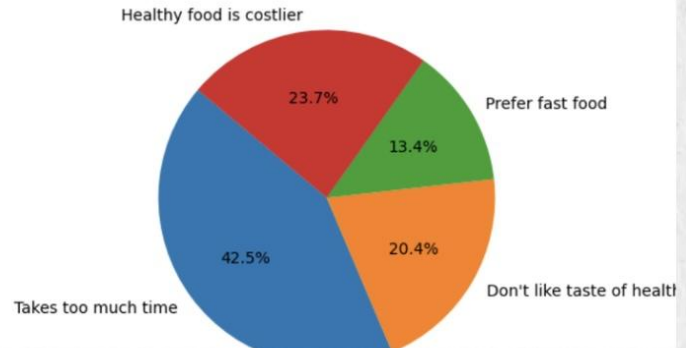
Frequency of ≥ 1 Hour Physical Activity per Week Among Wichita Adolescents



*While a **majority of adolescents** report **some physical activity**, a **significant proportion** do not meet **daily activity recommendations**, increasing **obesity risk**.*

What do Adolescents think about barriers for eating healthy

Perceived Barriers to Eating Healthy Among Adolescents (n=134)



***Time constraints** were the **most commonly reported barrier**, followed by **cost concerns**, **taste preferences**, and **preference for fast food**. These findings highlight how **convenience** and **affordability** strongly influence adolescents' dietary choices, particularly among **low-income populations**.*



Sustainability

Our solution is designed to be low-cost, community-embedded, and youth-led, allowing it to continue well beyond initial funding.

- **School & Community Integration**

Programs are embedded within existing school systems and community organizations, reducing reliance on new infrastructure.

- **Train-the-Youth Model**

Older students mentor and train younger peers, ensuring continuity as students graduate.

- **Reusable Toolkits & Curriculum**

Educational materials and program guides are developed once and reused across schools and communities.

- **Partnership-Driven Support**

Ongoing collaboration with schools, nonprofits, and local food access organizations sustains resources and reach.

- **Data-Driven Improvement**

Continuous data collection allows programs to adapt, demonstrate impact, and secure future support.

- **Scalable Without High Cost**

The model can expand to new schools with minimal additional funding, making it sustainable and replicable.





Funding Ask

Funding Request:

We are requesting **seed funding** to transition this project from research to **real-world implementation** through a pilot program and scalable model.

What This Allows:

- Moves the project from data and research to action
- Creates a tested, evidence-based model for addressing adolescent obesity
- Improves access to nutritious food and physical activity for low-income youth
- Builds a scalable and sustainable solution that can expand to new schools
- Empowers youth to lead long-term public health and equity initiatives





Our Vision

A future where every adolescent, regardless of income or zip code, has access to nutritious food, opportunities for physical activity, and the ability to build healthy habits early in life.

Closing

Why It Matters

- Health disparities begin in childhood and widen over time
- Early, preventive action reduces long-term health and economic costs
- Youth-centered solutions create programs that are relevant, trusted, and sustainable

Invitation

We invite Civic Innovation Lab to partner with us in **turning research into action**, empowering youth voices, and building a **scalable public health model** that creates lasting impact for communities.





Acknowledgement

- Dr. Brenes / DR. Woods / Dr. Hawley - WSU, Dr. Thyfault & Dr. Ptomey - UKMC, Dr. Sarwer - Temple university, professors for providing guidance throughout the research process.
- Student participants, whose anonymous responses made this study possible.
- Chris Ashbrook, Paul Druzinsky, Chris Bloomer (school advisors) for distributing the data survey.

Survey

Local Wichita data survey: Teen Obesity & Healthy food access Data Survey



**Thank
You!**

